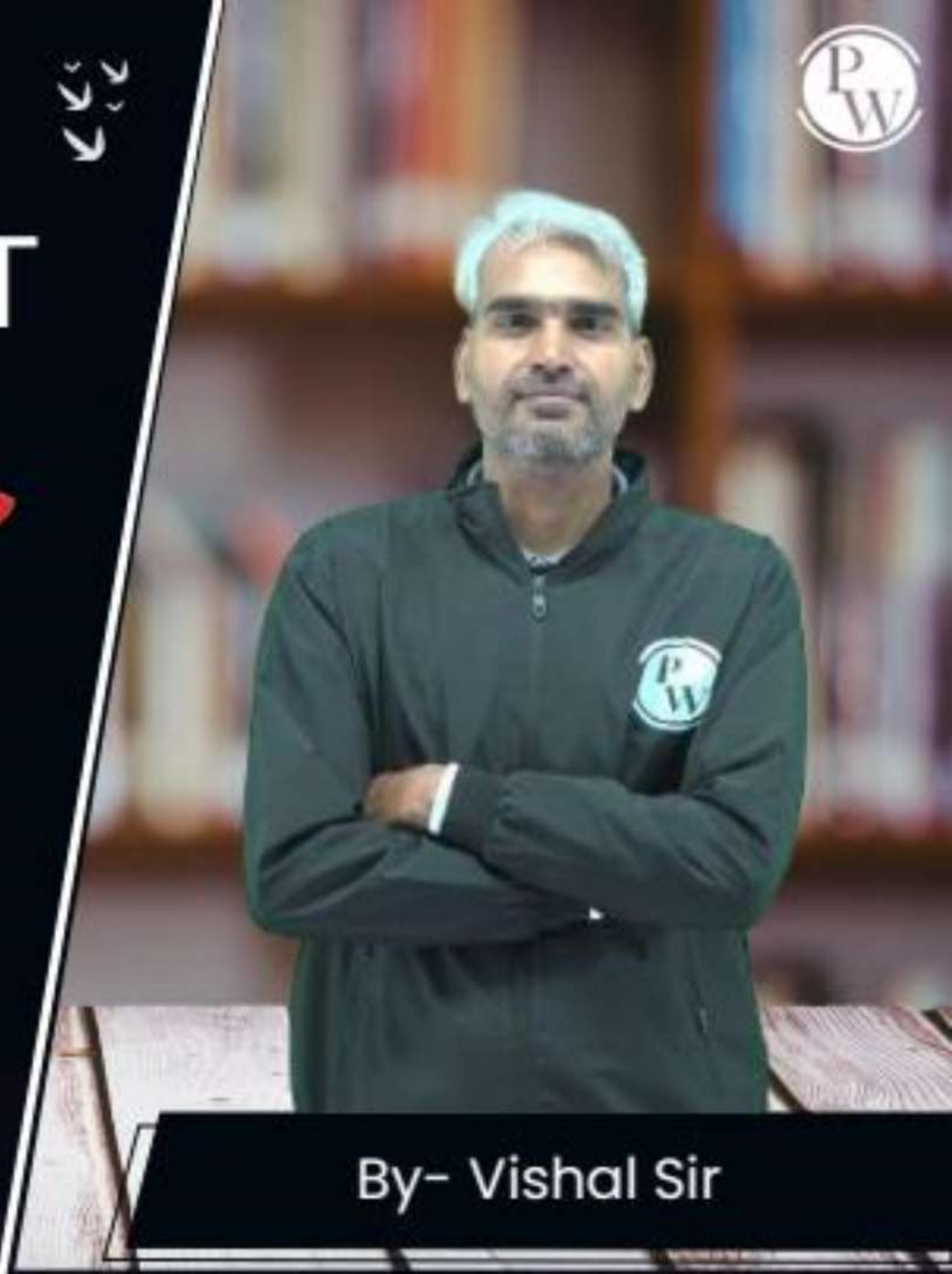
Computer Science & IT



Database Management System

Conversion from ER Model to Relational Model

Lecture No. 02



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Integrity constraints
- **Topic** Conversion from ER model to relational table



Slide

Topics to be Covered



Topic Conversion from ER model to relational table



Slide

ER Model to Relational Table

* ① one to one Relation

1



2



3



4



ER Model to Relational Table

(1) one to one Relation {Partial participation of both side entities}



P.K.

E1

A1	A2	A3
a1		
a2		
a3		

P.K.

R

A1	B1
a1	b1
a3	b3

P.K.

E2

B1	B2	B3
b1		
b2		
b3		

(E1 E2 R)

A1	A2	A3	B1	B2	B3
a1	–	–	b1	–	–
a2	–	–	NULL	NULL	NULL
a3	–	–	b3	–	–
NULL	NULL	NULL	b2	–	–

Primary key can not be defined for (E1 E2 R)
∴ Can not be merged into a single relation

P.K. F.K.

E1R

A1	A2	A3	B1
a1	–	–	b1
a2	–	–	NULL
a3	–	–	b3

P.K. F.K.

E2R

B1	B2	B3	A1
b1	–	–	a1
b2	–	–	NULL
b3	–	–	a3

Possibilities:

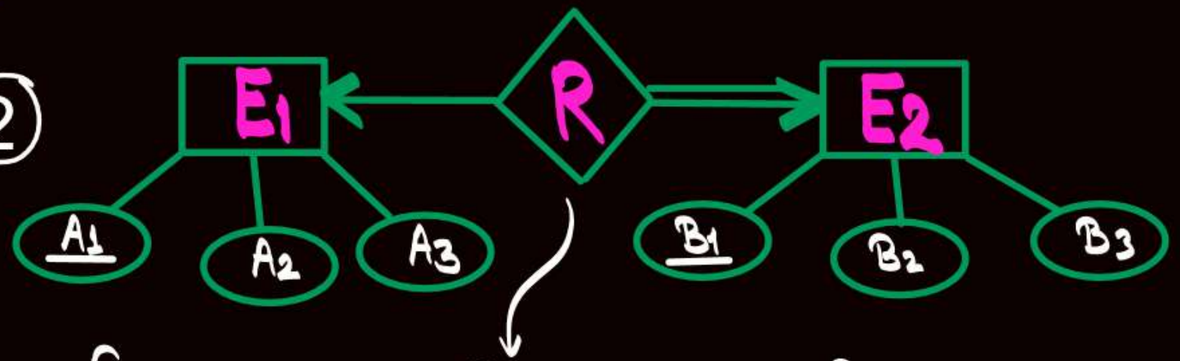
(1) E1R(A1, A2, A3, B1) and E2(B1, B2, B3)
P.K. P.K. F.K.

(or)

(2) E1(A1, A2, A3) and E2R(B1, B2, B3, A1)

ER Model to Relational Table

① one to one Relation { total participation of one entity set }

② 

E_1

A_1	A_2	A_3
a_1		
a_2		
a_3		

R

A_1	B_1
a_1	b_1
a_2	b_2

E_2

B_1	B_2	B_3
b_1		
b_2		

$(E_1\ E_2\ R)$

P.K.

A_1	A_2	A_3	B_1	B_2	B_3
a_1	–	–	b_1	–	–
a_2	–	–	b_2	–	–
a_3	–	–	NULL	NULL	NULL

{ We can Combine all three tables into a single table }

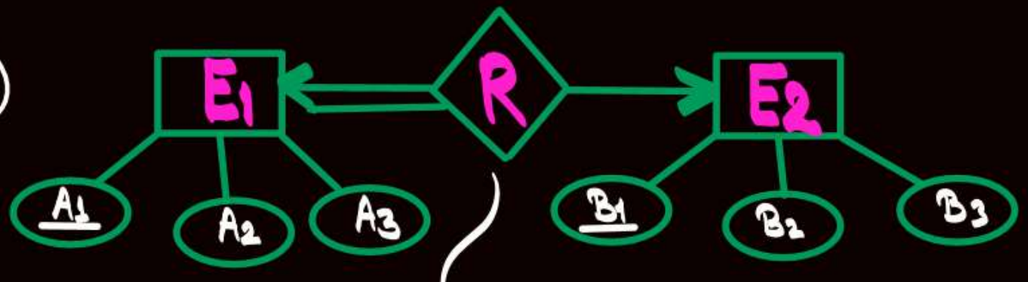
{ No foreign key }

Primary key = A_1 { Primary key of merged relation will be same as P.K. of Entity set with Partial participation }

ER Model to Relational Table

① One to one Relation { total participation of one entity set }

③



E_1

A_1	A_2	A_3
a_1		
a_2		

R

A_1	B_1
a_1	b_1
a_2	b_2

E_2

B_1	B_2	B_3
b_1		
b_2		
b_3		

$(E_1 E_2 R)$ — P.K.

A_1	A_2	A_3	B_1	B_2	B_3
a_1	–	–	b_1	–	–
a_2	–	–	b_2	–	–
NULL	NULL	NULL	b_3	–	–

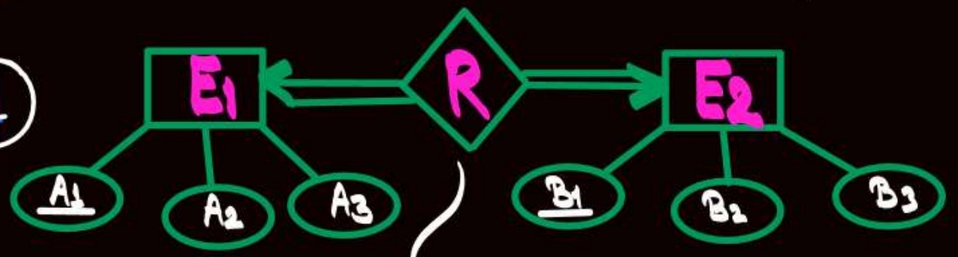
{ All three tables can be merged into single table

Primary key = B_1 { P.K. will be same as P.K. of Entity Set with partial participation }

ER Model to Relational Table

(1) one to one Relation {total participation of both side entity set}

(4)



E1

A1	A2	A3
a1		
a2		
a3		

R

A1	B1
a1	b1
a2	b2
a3	b3

E2

B1	B2	B3
b1		
b2		
b3		

E1 E2 R

A1	A2	A3	B1	B2	B3
a1	–	–	b1	–	–
a2	–	–	b2	–	–
a3	–	–	b3	–	–

All three tables can be merged into a single table (No foreign key)

Two candidate keys A1 and B1

Primary key = A1 or B1

ER Model to Relational Table

(1) one to one Relation

1



{ Minimum two tables are required }

2



{ only one table is required
P.K. = Same as P.K. of Entity set with Partial Participation }

3



{ only one table is required
P.K. = Same as P.K. of Entity set with Partial Participation }

4



{ only one table is required
P.K. : P.K of any side Entity set can be choosen as P.K of merged rel^n }

ER Model to Relational Table

② one to Many Relation

1



2



3



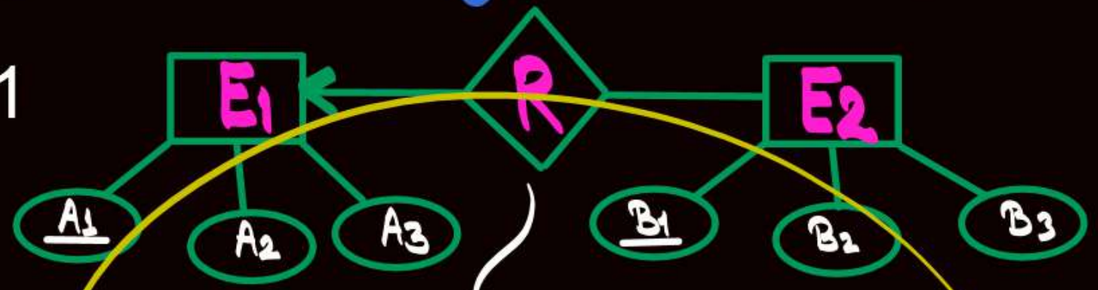
4



ER Model to Relational Table

② One to Many Relation { Partial participation of both side entities }

1



E1

A1	A2	A3
a1		
a2		
a3		

R

A1	B1
a1	b1
a1	b2

E2

B1	B2	B3
b1		
b2		
b3		

E1 E2 R

A1	A2	A3	B1	B2	B3
(E1R) can not be merged ∴ All three can not be merged into a single table					

duplicate

E1 R

A1	A2	A3	B1
a1	–	–	b1
a1	–	–	b2
a2	–	–	NULL
a3	–	–	NULL

P.k can not be defined ∴ R can not be NULL merged with E1

P.k

E2 R

B1	B2	B3	A1
b1	-	-	a1
b2	-	-	a1
b3	-	-	NULL

F.K.

Two tables are required

E1(A1, A2, A3) & E2R(B1, B2, B3, A1)

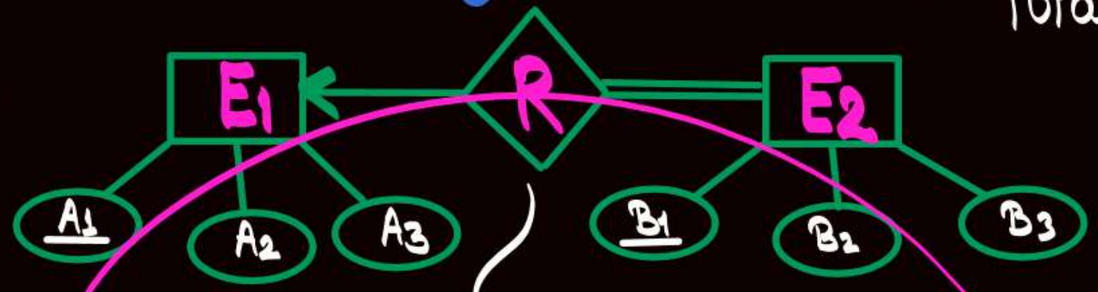
F.K

Relationship set will merge with many side entity set.

ER Model to Relational Table

② One to Many Relation { Partial Participation of '1' side entity set, Total Participation of 'Many' side entity set

2



All tree Can not be merged together

E_1

A_1	A_2	A_3
a_1		
a_2		
a_3		

R

A_1	B_1
a_1	b_1
a_1	b_2
a_2	b_3

E_2

B_1	B_2	B_3
b_1		
b_2		
b_3		

Two tables are required

$E_1(\underline{A_1}, A_2, A_3)$ & $E_2R(\underline{B_1}, B_2, B_3, A_1)$

F.K

duplicate

E_1R

A_1	A_2	A_3	B_1
a_1	—	—	b_1
a_1	—	—	b_2
a_2	—	—	b_3
a_3	—	—	NULL

No P.K.
∴ Can not be merged

NULL

P.K.

f.k

E_2R

B_1	B_2	B_3	A_1
b_1	-	-	a_1
b_2	-	-	a_1
b_3	-	-	a_2

Note:- * In one-to-many,

If participation of one side Entity set is partial, then minimum two tables are required, and relationship set will merge with many side entity set

* Primary key of merged relation will be same as Primary key of many side Entity set

ER Model to Relational Table

② One to Many Relation { Partial Participation of "many" side entity set, Total Participation of "1" sid entity set

3



E1

A1	A2	A3
a_1		
a_2		

R

A1	B1
a_1	b_1
a_1	b_2
a_2	b_3

E2

B1	B2	B3
b_1		
b_2		
b_3		
b_4		

E1 R

A1	A2	A3	B1
a_1	–	–	b_1
a_1	–	–	b_2
a_2	–	–	b_3

P.K

E2 R

B1	B2	B3	A1
b_1	–	–	a_1
b_2	–	–	a_1
b_3	–	–	a_2
b_4	–	–	Null

P.K

E1 E2 R

A1	A2	A3	B1	B2	B3
a_1	–	–	b_1	–	–
a_1	–	–	b_2	–	–
a_2	–	–	b_3	–	–
Null	Null	Null	b_4	–	–

All three tables can be merged into a single table (No foreign key)

$E_1E_2R(A_1, A_2, A_3, \underline{B_1}, B_2, B_3)$

P.K will be same as P.K. of many side Entity set.

ER Model to Relational Table

② One to Many Relation { Total participation of both entity set }

4



E1

A1	A2	A3
a1		
a2		

R

A1	B1
a1	b1
a1	b2
a2	b3

E2

B1	B2	B3
b1		
b2		
b3		

E1 E2 R

A1	A2	A3	B1 (P.K)	B2	B3
a1	–	–	b1	–	–
a1	–	–	b2	–	–
a2	–	–	b3	–	–

E1 E2 R (A1, A2, A3, B1, B2, B3)

{P.K is same as P.K of many side Entity set}

Note:- * In one-to-many,

If participation of one side Entity set is "total" then only one table is required, and

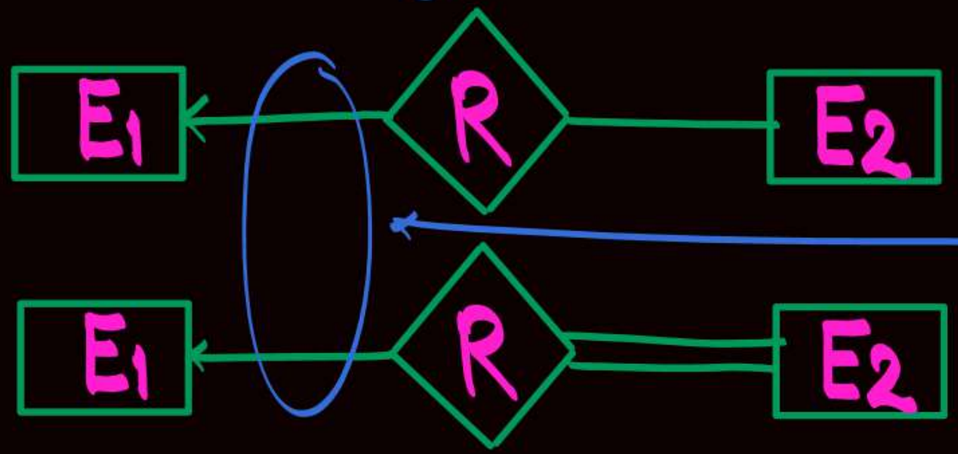
* Primary key of merged relation will be same as Primary key of many side Entity set

ER Model to Relational Table

② One to Many Relation

1

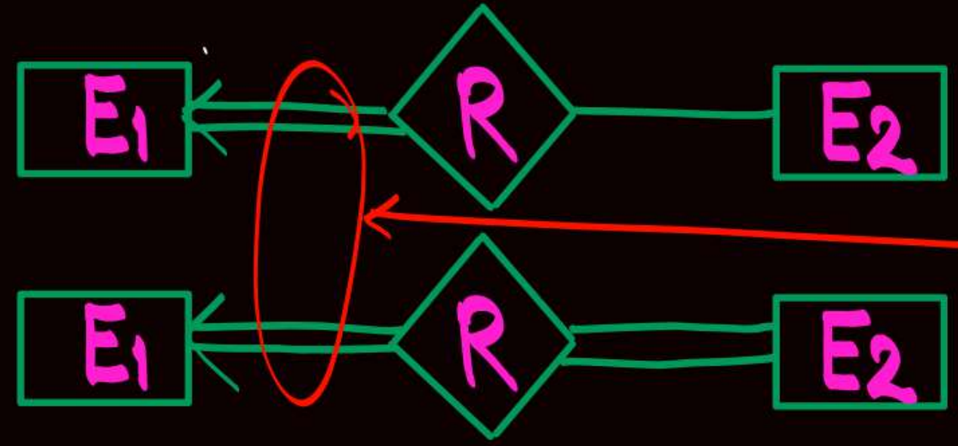
2



Partial Participation of "One" side Entity set.
∴ Two tables are required
→ Relationship set will merge with many side Entity set
Pk: Same as P.k of many side Entity set.

3

4



Total Participation of one side Entity set.
∴ only '1' table is required
P.k: Same as P.k of many side Entity set

ER Model to Relational Table

(3) Many to one Relation

{ Exactly same as One-to-many }

1



2



3



4



Partial Participation of one side Entity set
∴ Two tables are required

Total participation of one side Entity set
∴ Only '1' table is required

ER Model to Relational Table

(4) Many to Many Relation { Partial participation of both side entities }

1



E_1

A_1	A_2	A_3
a_1		
a_2		
a_3		

R

A_1	B_1
a_1	b_1
a_1	b_2
a_2	b_2

E_2

B_1	B_2	B_3
b_1		
b_2		
b_3		

duplicate

E_1R

A_1	A_2	A_3	B_1
a_1	–	–	b_1
a_1	–	–	b_2
a_2	–	–	b_2
a_3	–	–	NULL

No P.K. ∴ Can not be merged

NULL

duplicate

E_2R

B_1	B_2	B_3	A_1
b_1			a_1
b_2			a_1
b_2			a_2
b_3			NULL

No P.K. ∴ Can not be merged

NULL

* We will require minimum three tables in relational model

$E_1(\underline{A_1}, A_2, A_3)$ — P.K

$E_2(\underline{B_1}, B_2, B_3)$ — P.K

$R(\underline{A_1, B_1})$ — f.k, f.k

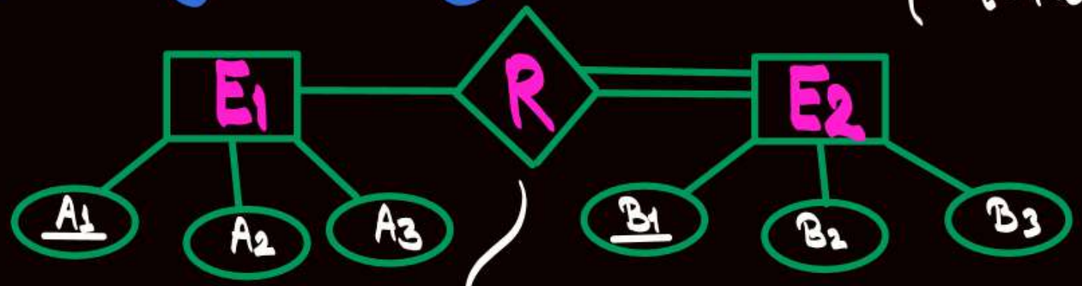
P.K. of R will be (A_1B_1) together

ER Model to Relational Table

(4) Many to Many Relation

{ Partial Participation at one side, total Participation at another side }

2



→ All three can not be merged into a single table

Two tables are required

$E_1(\underline{A_1}, A_2, A_3)$ and $E_2R(\underline{B_1, A_1}, B_2, B_3)$

P.K

P.K = $(B_1 A_1)$

F.K.

E_1

A_1	A_2	A_3
a_1		
a_2		
a_3		

R

A_1	B_1
a_1	b_1
a_1	b_2
a_1	b_3
a_2	b_3

E_2

B_1	B_2	B_3
b_1		
b_2		
b_3		

E_1R

A_1	A_2	A_3	B_1
a_1	—	—	b_1
a_1	—	—	b_2
a_1	—	—	b_3
a_2	—	—	b_3
a_3	—	—	NULL

not unique

No. P.K.

∴ Can not be merged

NULL

E_2R

B_1	B_2	B_3	A_1
b_1	—	—	a_1
b_2	—	—	a_1
b_3	—	—	a_1
b_3	—	—	a_2

Not unique

Together $(B_1 A_1)$ values will be unique

∴ P.K = $(B_1 A_1)$ together

not unique

Relationship set will Merge with the Entity Set with "total Participation"

ER Model to Relational Table

(4) Many to Many Relation

{ Partial Participation at one side, total Participation at another side }

3



E_1

A_1	A_2	A_3
a_1		
a_2		
a_3		

R

A_1	B_1
a_1	b_1
a_1	b_2
a_2	b_2
a_3	b_1

E_2

B_1	B_2	B_3
b_1		
b_2		
b_3		

Same as previous Case

Two tables are required

$E_2(\underline{B_1}, B_2, B_3)$ & $E_1R(\underline{A_1, B_1}, A_2, A_3)$

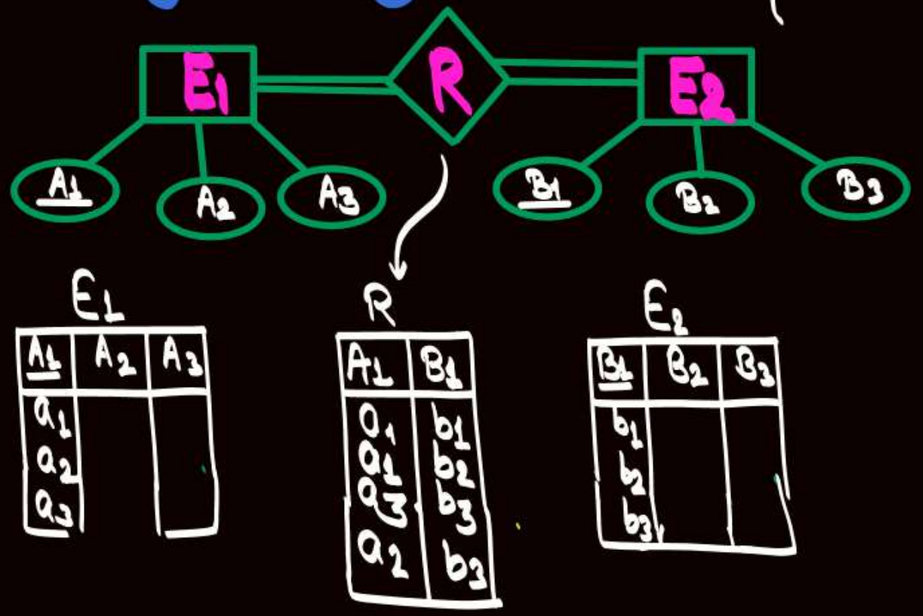
B_1: Pk

B_1: F.K

A_1, B_1: P.k = $(A_1 B_1)$

ER Model to Relational Table

(4) Many to Many Relation {Total participation at both side}



E1 E2 R

P.K = $(A_1 B_1)$

A_1	A_2	A_3	B_1	B_2	B_3
a_1	—	—	b_1	—	—
a_1	—	—	b_2	—	—
a_2	—	—	b_3	—	—
a_3	—	—	b_3	—	—

All three can be merged into a single table

ER Model to Relational Table

(4) Many to Many Relation

1 

{ Three table }

2 

{ Two tables } P.k. col Merged table (E2R) = (P.k. col E1 + P.k col E2)

3 

{ Two tables } P.k. col Merged table (E1R) = (P.k. col E1 + P.k col E2)

4 

{ One table } P.k. col Merged table (E1E2R) = (P.k. col E1 + P.k col E2)

ER model to relational model w.r.t. Week entity set



Relational table w.r.t Employee Entity set

Employee

Eid	Ename

Employee(Eid, Ename)

P.K.

identifying relationship set will always merge with weak Entity set after merging

Relational table

F.K.

Dependent_takescare (Eid, d_name, age)

Primary Key



2 mins Summary

Topic Conversion from ER model to relational table

Slide

THANK - YOU